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See also: Stoeckle MY, Soboleva L, Charlop-Powers Z. Aquatic environmental DNA detects seasonal fish abundance and habitat preference in an urban estuary. PLOS ONE 2017: e0175186 [link](#)

Overview. From start to finish there are four laboratory components, each of which can potentially be completed in 1 day. Except for filtration, which we aim to do within 0-2 days of collection, samples can be safely stored for weeks to months between steps at -20 °C. All procedures are done on an open bench wearing gloves; wipe down work area and equipment after use with 10% bleach. We change gloves a lot to minimize cross contamination. Separate sets of pipettors and reagents are used for pre- and post-PCR work. We do one component at a time; in particular when filtering do not perform other procedures including DNA isolation and PCR. Unfiltered pipette tips are employed; after each procedure used tips are discarded and collection containers rinsed with 10% bleach. To reduce potential for cross-contamination, we do PCR in GE Illustra 8-tube strips (i.e., not 96-well plates), and other steps in individual 1.5 ml microfuge tubes.

LABORATORY PROCEDURES

1. Filter water samples
2. Isolate DNA
 - A. Isolate DNA from filter
 - B. quantify DNA
3. Amplify vertebrate 12S
 - A. PCR vertebrate 12S
 - B. check PCR products by gel electrophoresis
 - C. AMPure purify or dilute PCR reaction mix
4. Add Nextera tags, pool libraries
 - A. PCR to add Nextera barcode tags
 - B. check PCR products by gel electrophoresis
 - C. purify PCR products
 - D. quantify PCR products
 - E. standardize library concentration, pool libraries
 - F. deliver pooled library sample to sequencing facility

1. Filtration water sample

Summary: Filter 1 liter water sample to concentrate aquatic eDNA

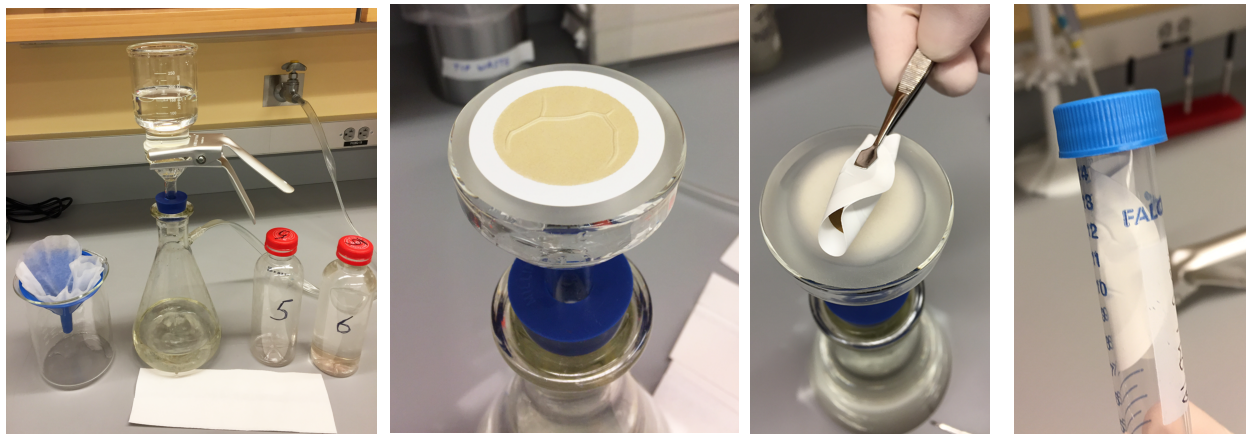
Time: About 2-3 h depending on how quickly filter clogs

Equipment: 1000 ml side arm flask
250 ml filtration manifold (Millipore)
Nylon filters: 47 mm diameter, 0.45 μ M pore size (Millipore)
Flat forceps for handling filters
Paper coffee filters
Plastic funnel

1. Store samples at 4 °C until filtered. We have filtered water up to 48 h after collection with good eDNA detection and up to 1 month with at least some recovery of eDNA. However, after 10 days only the most abundant eDNAs were still detectable.

2. Assemble filtration equipment listed above and attach to wall suction. Pour sample into manifold through a paper coffee filter placed in a funnel (this removes large debris). The volume that can be filtered before process slows drastically seems to correlate with the amount of DNA recovered per liter, so presumably slowly-filtering samples have a lot of planktonic organisms (bacteria, algae, other). It usually takes about 2-3 h to filter one liter from marine samples. For samples that filter very slowly, we usually stop the process after about 4 hours and record the volume filtered. We filter 1 liter samples of tap water as negative controls.

3. When filtration complete, disassemble apparatus, fold the filter over twice using forceps (sample side inward so it doesn't rub on side of tube), place in 15 ml tube, and store in -20C freezer. Filters have been stored up to 6 months with normal DNA yields and no obvious loss of eDNA detections. After filtration we rinse collection bottles thoroughly and scrub filtration equipment extensively with tap water, taking particular care to remove any material adherent to glass manifold by rubbing under tap with a paper towel.



2A. DNA Extraction from filter

Summary: Extract DNA using MoBio Powersoil Kit*

Time: About 2 h. We have processed up to 8 filters per person at a time.

Equipment: MoBio PowerSoil Kit (includes buffers C1-C6, PowerBead Tubes, Collection Tubes, Spin Filters)
Vortex platform with clips to hold PowerBead tubes
Single-edge razor
Forceps
Petri dish

*We had surplus PowerSoil kits; PowerWater Kit would likely be easier as they do not require cutting up filters. Other differences from manufacturer's protocol include the amounts of buffers C1, C2, C3, C4. This modified protocol has worked well so far.

1. Using a single-edge razor and plastic petri dish as a cutting surface, first trim filter, discarding the outer "unused" portion that was not exposed to the filtrate (this makes it easier to fit filter into one PowerBead tube. Then with filter folded in half in a plastic petri dish, cut filter into pieces approximately 2-3 mm x 10 mm, using single-edge razor. Use forceps (or the 15 ml tube the filter was stored in, turned upside down with cap off) to hold the filter while cutting. Place cut up filter into a PowerBead tube. Use a new petri dish for each filter; between filters change gloves, clean forceps and single edge razor under tap water using paper towels for mechanical cleaning. After 4 filters the razor becomes dull, use a new one.

2. Add 180 μ l C1 solution.

3. Place on vortexer platform on high for 10-15. Watch that device does not fall off bench. This is probably the most critical step in DNA yield; if mixing appears limited due to amount of filter, we flip the tubes after 5 and 10 min and extend the total time to 15 min.

4. Centrifuge at 10,000g for 1 min.

5. Transfer ~465 μ l supernatant to a 2 ml Collection Tube (1 Collection Tube per PowerBead Tube). We use a P200 set at 155 μ l, and transfer 3 aliquots of supernatant from each PowerBead tube. You can stick pipette tip down deeply along the tube wall, beads and paper fragments are too large to go into pipette.

6. Add 100 μ l of Solution C2 and 100 μ l of Solution C3. Vortex 10 seconds.

7. Incubate at 4° °C for 5 min (put rack in refrigerator/cold room).

8. Centrifuge at room temp 1 min at 10,000g.
9. Transfer 625 µl supernatant to a new 2 ml Collection Tube.
10. Shake Solution C4 before use. Add 1 ml Solution C4 to the supernatant. Invert 20 times to mix.
11.
 - Load 675 µl onto a Spin Filter and centrifuge at 10,000g for 1 min at room temp.
 - Discard flow through.
 - Add another 675 µl to Spin Filter and centrifuge at 10,000g for 1 min.
 - Discard flow through.
 - Load the remaining supernatant onto Spin Filter and centrifuge at 10,000g for 1 min.
 - Discard flow through.
12. Add 500 µl Solution C5 to Spin Filter and centrifuge at room temp for 30 secs at 10,000g. Discard flow through.
13. Centrifuge at room temp for 1 min at 10,000g.
14. Place Spin Filter in a clean 2 ml Collection Tube.
15. Add 50 µl Solution C6 to the center of the white filter (don't touch filter with pipette tip).
16. Centrifuge at room temp for 30 secs at 10,000g. Usually recover ~45 µl (compare to 50 µl C6 input). Store at -20 °C.
17. Measure DNA yield with Qubit, DNA yield from surface ocean water samples so far averages about 2000 ng per liter filtered (range 500 ng -10,000 ng)

2B. Quantify DNA: QuBit Protocol

Summary: Assay DNA concentration with QuBit (Qiagen)

Time: 15 min

Equipment: Qiagen Qubit machine
Qiagen High Sensitivity (HS) dilution buffer, HS dye
Qiagen HS DNA standards

1. Set out 0.25 ml PCR tubes in tray (need $n+2$, where n = number of samples).
2. Make buffer-dye mix
Qiagen HS buffer solution: $[(n+3)*200 \mu\text{l}]$. Add $(n + 3) \mu\text{l}$ of HS dye, vortex to mix.

2. Add standards, samples to PCR tubes
Tube 1: 10 μl of HS Standard 1
Tube 2: 10 μl of HS Standard 2
Sample tubes: 2 μl per sample

3. Add dye mix
Standard Tubes 1,2: Add 190 μl buffer-dye mix
Sample tubes: Add 198 μl buffer-dye mix

4. Measure DNA concentration (select dsDNA, High Sensitivity):
Read Standard Tubes 1,2
Read sample tubes (note Qubit reading is in ng/ml)
Calculate sample concentration
Note: sample concentration in $\text{ng}/\mu\text{l}$ = Qubit reading (which is in ng/ml) $\times 1/10$

3A. PCR amplification of vertebrate 12S fragment

Summary: Amplification of vertebrate 12S DNA from eDNA sample
12S primers from Riaz Nucl Acids Res 2011 39: e145)
Amplification strategy based on Illumina 16S protocol

Time: About 2.5 h to set up and run PCR, another 1 h to run gel to visualize results.

Equipment: GE Illustra beads (8 tube strips)
Thermal cycler
DNA primers (12S with Illumina tails)

Forward: 5'- TCG TCG GCA GCG TCA GAT GTG TAT AAG AGA CAG ACT GGG ATT
AGA TAC CCC -3'

Reverse: 5'- GTC TCG TGG GCT CGG AGA TGT GTA TAA GAG ACA GTA GAA CAG
GCT CCT CTA G -3'

Agarose gel electrophoresis equipment

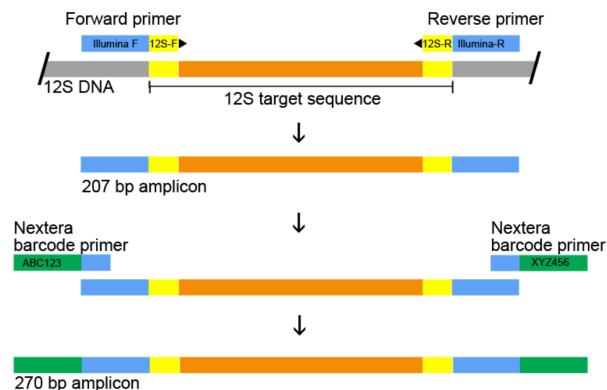
SyberSafe dye

1. Prepare 2 μ M primer stock solution in Elution Buffer (2 μ M each primer)
2. Make Master Mix (17.5 μ l H₂O + 2.5 μ l primer stock solution) x (number of PCRs + 1)
3. Spin GE Illustra PCR tubes briefly to insure beads are at bottom
4. Add following to PCR tubes
20 μ l Master Mix (final concentration of each primer is 200 nM)
5 μ l DNA (represents DNA from 100 ml of water) or 5 μ l H₂O for negative control

Spin briefly to make sure all reagents including bead are at bottom of tube.

3. Amplification protocol: 95 x 5m
40 cycles: 95 °C x 20s, 52 °C x 20s, 72 °C x 20s
72 °C x 1m, then hold at 4C

Schematic of amplification strategy

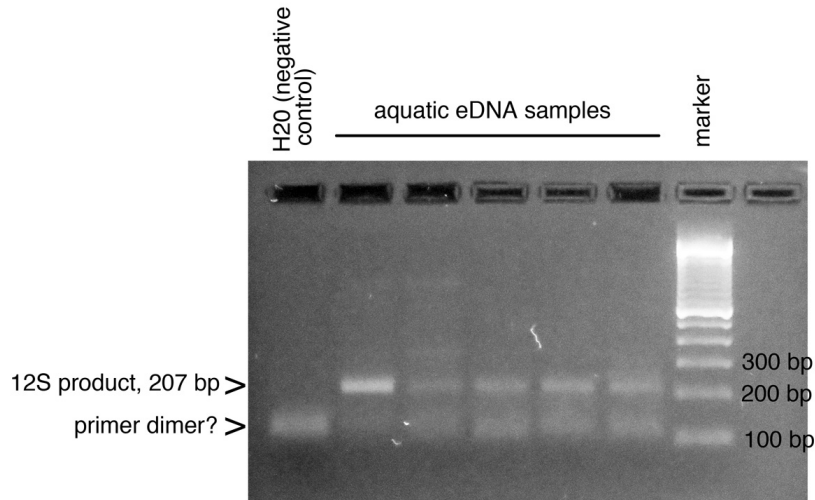


3B. Check PCR yield

1. Run 5 µl each PCR tube on 2.5% agarose gel in 1X TBE with SyberSafe dye and DNA size markers. Expected product approximately 207 bp. We also usually get a band at about 100 bp including in negative control; this presumably is a primer dimer.

In some runs there is a noticeable 200 bp band in negative controls. So far these have turned out to be human or domestic animal (pig, chicken, turkey, cow) amplicons. Human and domestic animal DNA are common contaminants in PCR reagents.

Sample gel:



3C. Purify PCR products—two alternative methods

Follow AMPure protocol (see below), resuspend in 40 µl Elution Buffer.

OR

An easier approach that works well so far is to simply dilute the PCR reaction mixes 1:20 in Elution Buffer (Qiagen).

DNA purification with AMPure beads

Summary: DNA purification, concentration using magnetic beads (this closely follows manufacturer's protocol)

Time: About 1-2 h, depending on number of samples.

Equipment: AMPure beads in solution per manufacturer's protocol
MagnaRack
Elution Buffer (Qiagen) or equivalent (10mM Tris, pH 8.3)
80% EtOH

1. Resuspend AMPure beads by inverting multiple times or vortexing.
2. Add 2:1 ratio by volume of AMPure beads to DNA sample. Place AMPure bead suspension into bottom of tubes, pipette 2-3 times to mix. If starting with PCR products, transfer rxn mixtures to bottom of 1.5 ml eppendorf tubes, then add AMPure beads.
3. Incubate at room temp 10 min.
4. Open tubes and place on MagnaRack for 5 min. Solution should become clear, with beads along back wall.
5. Carefully remove supernatant with P200 pipette, avoid touching beads.
6. Add 500 μ l 80% ETOH while tubes are on MagnaRack (gently pipet in ETOH along the side of the tube, do not re-suspend beads). Per Illumina protocol, minimum wash volume is (DNA sample + bead) volume.
7. Carefully remove supernatant with pipette. Do not remove tubes from rack.
8. Repeat EtOH wash (steps 6 +7). Do not remove tubes from rack. After aspirating second wash, use a P20 pipette to remove any remaining ETOH.
9. Let tubes air dry in rack for 5 min. Beads are hard to re-suspend after a long drying time.
10. Disassemble MagnaRack so tubes are not adjacent to magnets. Add desired volume Elution Buffer (EB) to each tube (we re-suspend eDNAs in 54 μ l and PCR products in 44 μ l; as noted below, the extra 4 μ l are to minimize bead transfer at final step), then mix by vortexing until beads are completely resuspended.
11. Incubate room temperature 5 min.
12. Assemble MagnaRack so tubes are adjacent to magnets. Before assembly, open the caps. Wait 2 min. Solution should be clear.
13. Without removing tubes from rack, carefully pipette the supernatant (whatever volume of Elution Buffer was added in step 10 less 4 μ l), transfer to a new tube, store at -20C. (Leaving behind 4 μ l reduces chance of beads in final sample. [note: per Illumina protocol, carry-over of a small amount of beads usually does not interfere with subsequent reactions.]

4A. Add Nextera tags-Adapted from Illumina 16S protocol

GE Illustra tubes (25 µl rxn volume)

-Input DNA: 5 µl of AMPure-purified/or diluted 12S PCR product

-Nextera tags: 2.5 µl each index (forward and reverse)

15 µl H₂O

PCR conditions 95 °C x 5m

[95 °C x 20s, 55 °C x 20s, 72 °C x 20s] x 10 cycles

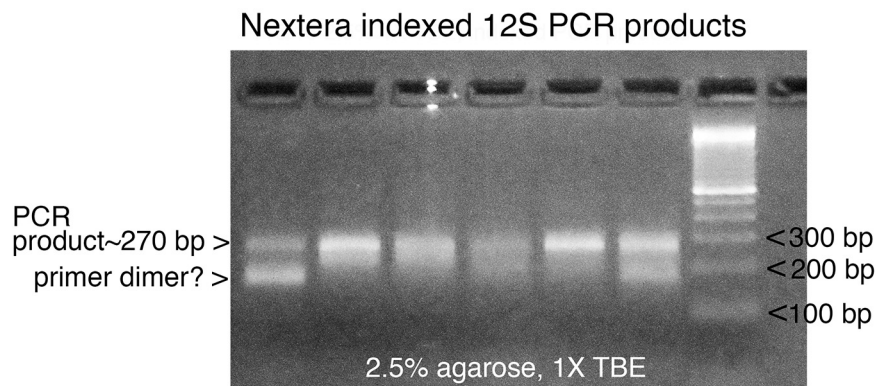
72 °C x 1m, hold at 4 °C

If diluted PCR product is input, use 13 cycles

4B. Check product size by gel electrophoresis

-Run 5 µl each sample on 2.0% agarose 1X TBE, with SyberSafe dye, markers

-Expected size about 270 bp



4C. Purify PCR products (AMPure)*

-Resuspend in 40 µl Elution Buffer

4D. Quantify PCR products (QuBit)*

4E. Prepare pooled library, deliver to sequencing facility

-Prepare 30 nM pooled library in Elution Buffer

-If product ~270 bp, then 30 nM = 5.4 ng/µl

-Add 5.4 ng x 20 = 108 ng each library to 1.5 ml eppendorf

-Then add Elution buffer to final volume (#libraries x 20 µl)

-e.g., if 10 libraries, then final volume is 200 µl

*An alternative (that is much less work!) to purifying and quantifying each library separately is to simply add the same volume (e.g., 10 µl) of each library to a pooled sample, AMPure purify, then measure the DNA concentration of the pool, and adjust the concentration to 15 nM as described above. Whether this is appropriate depends on the nature of your experiment. In some settings, this equal-volume approach may produce a pool that is more representative of the differences among samples.